

Systematic Literature Review of Vulnerabilities and Defenses in VPNs, Tor, and Web Browsers

Neha Agarwal^{*[0009-0009-8740-2419]}, Ethan Mackin^{*[0009-0002-8462-1906]}, Faiza Tazi^{*[0000-0001-9781-2694]}, Mayank Grover^{†[0009-0004-9524-8427]}, Rutuja More^{†[0009-0006-5941-8294]}, and Sanchari Das^{†[0000-0003-1299-7867]}

^{*}University of Denver, USA

{Neha.Agarwal,Ethan.Mackin,faiza.tazi}@du.edu

[†]George Mason University, USA

{mgrover3,rmore,sdas35}@gmu.edu

Abstract. Modern web apps deliver richer experiences, including interactivity, dynamic content, and real-time communication, but their complexity widens the attack surface, enabling phishing, malicious extensions, memory-resident malware, and browser integrity attacks. These risks intensify with privacy tools like VPNs and Tor, whose protections can degrade or be bypassed under real-world adversaries. To map these vulnerabilities, we analyzed the architectures of VPNs, Tor, and mainstream browsers, identifying shared threat models, attack vectors, and defenses. From 552 articles, we selected 19 peer-reviewed studies that met strict criteria. They proposed defenses such as signature-based malware detection, secure extension APIs with script isolation, partitioned browser architectures, and dynamic blacklisting. Notably, 47% of the studies focused on VPNs or Tor, and 32% examined how these tools interact, both cooperatively and in conflict, in practice. Our results reveal gaps in how layered privacy technologies are deployed and understood; combining tools without aligning their assumptions fosters false confidence. We highlight automated and user-driven mitigations that, when integrated at the architectural layer, can meaningfully reduce privacy and integrity risks in routine web use.

Keywords: Web Browser · Tor · VPN · Security Analysis.

1 Introduction

The architecture of web has evolved dramatically since Berners-Lee’s 1990 HTTP client-server system [1]: Mosaic’s 1993 release enabled graphical, multimedia pages and fueled rapid growth [2–4], while CGI and JavaScript introduced dynamic, client-side interactivity that set the stage for Web 2.0 [5]. Today’s browsers are multi-process platforms that execute untrusted code (JavaScript, WebAssembly) [6–9], access sensitive resources, and support high-bandwidth APIs like WebRTC [10], relying on the same-origin policy, CSPs, and sandboxing for security [11]. As they have grown into full execution environments, their attack surface has expanded: attackers leverage drive-by downloads [12], extension

APIs [13], and residual memory states [14]. Related work on Tor environments underscores persistent user-facing privacy and anonymity gaps [15], highlighting ongoing risks to confidentiality, integrity, and availability.

To address the inherent limitations of browser-native security models, users often deploy network-layer privacy tools such as Virtual Private Networks (VPNs) and anonymity networks like Tor. VPNs offer encrypted tunnels over insecure networks and are commonly used to protect data-in-transit, especially on untrusted infrastructures [16]. Despite their popularity, VPNs have been shown to leak metadata and facilitate user tracking under various adversarial models [17]. Meanwhile, the Tor network employs layered encryption and onion routing to achieve anonymity, distributing trust across a decentralized volunteer-operated infrastructure [18, 19]. Tor is designed to defend against network surveillance and traffic correlation, yet recent work highlights its susceptibility to memory forensics [20], active deanonymization attacks, and denial-of-service (DoS) vectors [21]. In this work, we conduct a structured literature review to assess the security and privacy guarantees of three key pillars of online interaction: web browsers, VPNs, and the Tor network. Specifically, we analyze architectural vulnerabilities, attack surfaces, and adversarial threat models that undermine these technologies. We also survey mitigation techniques proposed in recent academic and industry research, categorizing them based on scope, deployment feasibility, and resilience under realistic attack scenarios [22]. Our contributions are:

- We provide a consolidated taxonomy of vulnerabilities affecting mainstream browsers, VPN protocols, and Tor, with emphasis on their operational and architectural causes.
- We evaluate proposed defenses, including code partitioning, signcryption schemes, proxy aggregation, and DOM integrity frameworks assessing their effectiveness and trade-offs.
- We discuss systemic limitations in the current ecosystem and propose directions for future research in secure browser design, privacy-preserving network protocols, and user-centric threat modeling.

2 Method

We conducted a systematic literature review, using the structured study designs from our prior surveys [15, 23–37] to examine security vulnerabilities and mitigation strategies in three widely deployed privacy-enhancing technologies: web browsers, VPNs, and the Tor anonymity network [16, 38–40]. Our objective was to synthesize peer-reviewed evidence on critical system-level flaws, identify the adversarial models under which these technologies are most vulnerable, and assess the robustness and practicality of proposed defenses [38, 39]. To ensure transparency and reproducibility, we followed a structured review protocol, summarized in the PRISMA-style flow diagram shown in Figure 1 [16]. This framework guided our identification, screening, and inclusion process and helped mitigate bias during corpus construction and analysis [16]. This review aimed to answer two central research questions (RQs):

- RQ1: Where do existing web browsers fall short in safeguarding user data, and what technical proposals have emerged to address these weaknesses?
- RQ2: What security and anonymity limitations persist in Tor and VPN-based systems, and how effective are the available mitigation strategies under realistic threat models?

2.1 Paper Collection

We searched January 2010–March 2025 across Google Scholar (primary) and IEEE Xplore, SpringerLink, and ACM Digital Library, focusing on modern browser architectures, major VPN protocols (OpenVPN, IPsec) [16, 41, 42], and the Tor network [18, 43]. Boolean queries combined “VPN security,” “VPN vulnerabilities,” “web browser security,” “web extensions and malware,” and “Tor security/vulnerabilities” [38, 39, 44]. We included peer-reviewed publications written in English language.

The initial search yielded 552 papers. After removing 47 duplicates, 505 remained for title and abstract screening. We excluded 385 that lacked system-level analysis of browser, VPN, or Tor security—most focused on policy, ethics, or UI design without technical evaluation [16, 45, 46], or presented broad overviews without specific vulnerabilities or countermeasures [47, 48]. This left 120 papers for full-text review. Thereafter, we retained studies meeting three criteria: (i) analysis of specific technical vulnerabilities in browsers, VPNs, or Tor [40, 49, 50]; (ii) proposal or evaluation of a system-level defense or mitigation [39, 44]; and (iii) inclusion of empirical, implementation-based, or systematic analysis rather than conceptual discussion [38, 45]. This process excluded 101 papers, primarily those lacking reproducible or applied results [16]. Finally, 19 peer-reviewed studies were included for analysis.

2.2 Analytical Approach

We performed a structured thematic analysis [38–40] of the final corpus of 19 studies to extract, normalize, and compare their core technical contributions. To ensure consistency and replicability in our synthesis, we developed a standardized annotation schema and applied it manually to each paper. For every study, we recorded the target technology domain (browser, VPN, or Tor) [41, 43, 44], the specific system components under analysis (for example, protocol layers, user interface modules, relay nodes) [10, 40, 51], and the class of vulnerability addressed (such as phishing attacks, memory leakage, traffic correlation, or authentication bypass) [40, 52]. We also classified each study according to its assumed adversarial model, distinguishing among local attackers, remote adversaries, man-in-the-middle (MitM) actors, compromised infrastructure nodes, and passive network observers [40, 41, 53]. We documented the type of mitigation strategy proposed or evaluated, categorizing them as architectural changes, protocol-level defenses, runtime hardening techniques, or detection mechanisms [44, 54]. In parallel, we examined the evaluation methodology employed by each paper, including formal verification, experimental simulation, proof-of-concept implementations, and

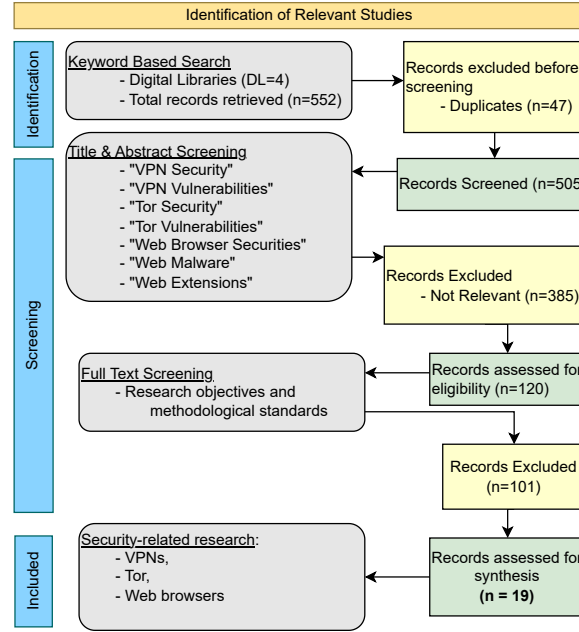


Fig. 1. PRISMA flow diagram illustrating identification, screening, eligibility assessment, and final inclusion of reviewed studies.

post-mortem forensic analysis [39, 40, 55]. Where applicable, we note whether the proposed solutions were system-specific (for example, Tor-only, browser-specific) or generalizable across multiple platforms [5, 18, 56].

After completing the individual annotations, we conducted an inductive thematic synthesis to identify patterns and categorize the studies by technological focus, namely, web browsers, VPNs, and the Tor network [44, 50, 56][27, 33, 35]. Within each category, we compared common threat vectors, adversarial assumptions, and defense models to reveal cross-cutting security challenges. These included persistent risks such as malicious browser extensions [5, 44], unbounded memory residue [12, 17, 55], phishing techniques [38, 52, 57], and insecure plugin APIs [5]. We also identified technology-specific weaknesses, including circuit deanonymization in Tor [40, 58], credential leakage in VPN tunnels [41, 42], and sandbox violations in browser architectures [12, 18]. This analytical approach enabled us to construct a comparative security landscape across the three systems and to evaluate the breadth, novelty, and practical feasibility of the defense mechanisms proposed in the literature. The resulting insights form the foundation of the findings presented in Section 3.

3 Findings

3.1 Vulnerabilities

We organize vulnerabilities into two domains: those affecting VPN and Tor systems, and those affecting web browsers.

VPN: These are widely adopted tools to enhance online privacy and data security by encrypting user traffic and masking IP addresses through remote servers. Despite their popularity, VPNs are not impervious to attack. These include misconfigurations, weak encryption standards, or insecure logging practices can expose users to network surveillance and data leakage.

Username Enumeration: VPN infrastructures using IPsec with Internet Key Exchange version 1 (IKEv1) in Aggressive Mode are vulnerable to username enumeration [41, 59]. Unlike IKE Main Mode, which encrypts identity payloads and hashes session keys, aggressive mode exposes the responder’s identity and a hash of the pre-shared key in plaintext. Attackers can replay handshake messages and verify guessed usernames offline. Since a common shared secret is often used by all clients, both confidentiality and identity privacy are compromised. VPN servers lacking rate limiting or anomaly detection remain open to large-scale automated enumeration. Virvilis et al. [60] further show that browser-integrated VPN authentication can leak identity hints through weak cross-origin protection or server validation. Thus, even encrypted channels may expose authentication data when protocol layering or endpoint validation is improperly enforced.

Link State Advertisement (LSA) Spoofing: VPNs using the Open Shortest Path First (OSPF) protocol are exposed to route manipulation via falsified LSAs [61]. Attackers can inject counterfeit LSAs, corrupt link-state databases, and alter routing to reroute, drop, or flood traffic. Because OSPF lacks authentication, adversaries can impersonate routers and modify topology data, disrupting forwarding and enabling interception or DoS. This shows that even with secure tunnels, unprotected control-plane protocols can compromise VPN availability.

Adjacent Route Spoofing (Path Spoofing): A related OSPF weakness arises when attackers exploit routers not set as passive interfaces to form fake adjacencies [61]. By initiating OSPF “hello” messages, adversaries gain neighbor data and inject malicious routing updates or LSAs. These forged adjacencies let them redirect traffic or split internal subnets. As many VPN gateways depend on dynamic OSPF routing, such spoofing undermines network integrity and can precede large-scale DoS attacks.

Resource Exhaustion via Diffie-Hellman Key Exchanges: IPSec VPNs using IKE in Main or Aggressive modes face CPU exhaustion from floods of spoofed initiation requests [61]. Each request triggers costly Diffie-Hellman computations, overloading servers and disconnecting clients. The attack executes faster in Aggressive Mode due to fewer handshakes, highlighting IKE’s asymmetric workload and lack of rate-limiting protections.

Tor: The Tor networks are designed to provide stronger anonymity guarantees by routing traffic through multiple volunteer-operated relays. While Tor

effectively conceals user identities against many forms of network surveillance, it too faces practical vulnerabilities [62, 63].

Memory Artifacts in Tor: Forensic studies have uncovered residual artifacts in both disk and memory [20, 43, 52]. These include IP addresses, circuit metadata, session keys, and visited onion sites. Jadoon et al. [20] classify them as disk artifacts (cache, swap), browsing traces (tabs, history), and Tor-specific items (identity keys, node descriptors). Critically, recovered onion routing keys may let attackers reconstruct circuits or decrypt sessions, contradicting assumptions of total data erasure. Chetry et al. [52] further show that such remnants can persist after normal shutdowns, depending on OS memory handling. Thus, Tor’s anonymity depends on both network design and host hygiene. Without memory sanitization and ephemeral keys, even short sessions can leave forensic traces.

DDoS Attacks: While Tor resists traffic analysis, its earlier architectures - especially for onion services - remain prone to DDoS attacks [21, 51, 54]. Adversaries can register rogue onion routers (ORs) via directory authorities, then flood onion services with connection or relay requests. Because each Tor request involves layered encryption and routing, resource exhaustion quickly degrades performance and availability. Feng and Zhao [21] propose identity-based sign-encryption to harden node authentication. Their scheme replaces centralized directory checks with mutual node verification, where each router sends a combined signing-encryption value to peers. This decentralized process blocks unverified nodes, lowers handshake overhead, and prevents key reuse or impersonation. Peer-level authentication also curbs Sybil-based flooding and improves latency, showing that even privacy-preserving systems like Tor need cryptographically enforced admission controls to maintain service resilience.

Web Browsers: Browsers are the primary interface to the internet, but growing complexity and extensibility broaden their attack surface, enabling phishing, malware injection, insecure extensions, memory-safety bugs, and flawed content/script isolation. Our synthesis maps structural and behavioral weaknesses and evaluates the efficacy and deployment challenges of browser-level defenses.

Table 1: Security Vulnerabilities : VPNs, Tor, and Web Browsers

Technology	Vulnerability	Description
VPN (IKEv1 Aggressive Mode)	Username Enumeration	Leaked hashed identities enable offline brute-force of shared usernames and automated probing due to missing rate limiting. [41, 60].
VPN (IKE - main / aggressive modes)	Resource exhaustion	Flood of spoofed IKE requests triggers repeated key exchanges, overloading CPU and disrupting IPSec services.
VPN (OSPF)	Link State Advertisement (LSA) spoofing	Injected fake LSAs corrupt routing tables, rerouting or flooding traffic to cause large-scale VPN disruption.

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Technology	Vulnerability	Description
VPN (OSPF)	Adjacent Route Spoofing	Fake OSPF adjacencies allow attackers to insert malicious routes, enabling traffic redirection and DoS setups
Tor	Memory Artifacts	Forensic recovery of IPs, circuit paths, and cryptographic keys from RAM and disk after session ends. Undermines anonymity guarantees [20, 43, 52].
Tor	DDoS on Onion Services	Malicious relays can overload onion services by relaying high volumes of encrypted requests, leading to denial-of-service [21, 51, 54].
Web Browsers	Phishing and Drive-by Malware	Malicious websites use social engineering or exploit browser plugins to install malware silently. Blacklists offer insufficient protection [12, 57, 60, 64].
Web Browsers	Residue Objects	Improper object deallocation across pages allows memory corruption, visual spoofing, and stale cross-origin references [14, 40, 42, 65].
Web Browsers	Secure Connection Risks (WebRTC)	JavaScript-controlled media and poor peer validation can leak audio/video streams. Persistent keys enable session correlation [10, 66].
Web Browsers	Malicious Extensions	Poorly permissioned or outdated extensions inject code, leak credentials, or hijack sessions. Vetting and updates often inadequate [10, 13, 44].
Web Browsers	Monolithic Architecture (Web 2.0)	Shared fault domains allow code execution across tabs or applications. Lacks proper isolation for complex modern tasks [50].

Phishing and Malware: Social engineering and drive-by downloads exploit user trust and browser execution environments to deliver malicious payloads [12, 57, 64]. In drive-by attacks, obfuscated JavaScript or exploit kits execute silently upon visiting compromised sites, probing client configurations and triggering vulnerabilities. Social engineering manipulates users through deceptive interfaces or spoofed content, often bypassing technical safeguards. Blacklist-based defenses offer limited protection; Chang et al. [12] found inconsistent malware blocking rates across major browsers and frequent false negatives, while Virvilis et al. [60] observed even weaker defenses in mobile browsers due to infrequent updates and opt-in protection models.

To enhance detection, hybrid systems combine heuristics, anomaly detection, and behavioral analysis. *HoneyMonkey* [12, 67] emulates user browsing in virtualized environments to detect unauthorized system changes, while *JSAND* [12] uses machine learning to flag suspicious JavaScript patterns. These dynamic approaches outperform static blacklists in identifying evasive payloads and zero-day exploits. Yet, browser warnings remain a major weakness—Akhawe and Felt [38] found that over 70% of users ignore SSL alerts and phishing warnings. This highlights the need for context-aware notifications and integrated, real-time filtering at the browser engine level [68]. Technical controls alone are insufficient without stronger user-centered design and platform-wide support for proactive detection.

Browser Integrity: Web browsers suffer from residual object vulnerabilities where remnants of prior pages persist across navigations, exposing users to security risks [14, 40]. These residual elements—HTML objects, JavaScript references, or styles—may not be cleared when new documents load, especially if navigation is interrupted. The result is dangling references that leave obsolete memory objects accessible within the browser’s internal model [65]. Such artifacts compromise memory safety, document integrity, and visual consistency. Attackers can exploit them for visual spoofing, reusing trusted UI elements from legitimate sites (e.g., logos or buttons) to deceive users. Flaws in cross-origin enforcement may further allow untrusted scripts to access stale objects, undermining the same-origin policy and enabling code injection or privilege escalation [42].

Residue vulnerabilities range from minor UI issues to severe memory corruption, where partially freed objects enable code execution outside the browser sandbox. Attackers can thus alter browser state or access host resources, endangering privacy and system integrity. Mitigation needs memory-safe languages, modular sandboxing, and systematic object model analysis [14], yet legacy engine complexity and dynamic content make complete prevention difficult.

Secure Connections: Trust in client–server communication is vital for confidentiality and integrity, especially in real-time protocols like WebRTC. Unlike HTTPS, which relies on TLS, WebRTC establishes peer-to-peer links via JavaScript-based signaling servers [10]. While this improves flexibility, it introduces risks: compromised sites can misuse camera or microphone access [66], and weak signaling may enable impersonation or man-in-the-middle attacks. Reusing persistent encryption keys can further link user sessions [69]. Despite browser prompts and sandboxing, these safeguards often fail against social engineering or persistently granted permissions [70, 71]. Secure connections require robust authentication, frequent key rotation with forward secrecy, strict validation, and minimal client-side state to prevent surveillance and active exploitation.

Extensions and Plugins: Browser extensions enhance functionality but pose serious security risks. Many are poorly vetted, especially in third-party repositories, and often bypass permission prompts to access sensitive APIs and user data [44]. Malicious extensions can inject scripts, steal credentials, or redirect traffic, while trusted ones may later deliver harmful updates through supply-chain attacks. Although most browsers implement extension signing, obfuscated or unsigned code still evades detection [13].

Cross-browser analyses reveal inconsistent enforcement of security controls [10, 72, 73]. Firefox provides stronger permission models, whereas Safari and others lack key protections or automatic integrity checks. The WebExtension model compounds these risks by granting high privileges to background and content scripts; without strict isolation, compromised extensions can subvert browser trust. Toreini et al. [13] proposed *DOMtegrity*, a cryptographic framework that validates DOM integrity against malicious plugins through isolated JavaScript contexts and secure client-server verification. Stronger privilege separation, mandatory signing, behavior monitoring, and automatic patching remain essential to reducing the attack surface of modern browser extensibility [74]. Table 1 provides a summary of all vulnerabilities in the target technology domains.

3.2 Mitigation Strategies

Here we summarize defense strategies for VPNs, Tor, and browsers, emphasizing deployable solutions with proven security benefits.

VPNs: Although VPNs are designed to secure communication through encrypted tunnels, practical deployments reveal persistent weaknesses in authentication, routing, and key exchange mechanisms. Misconfigurations in IPsec, unverified routing updates, and limited rate control often expose VPN servers to denial-of-service and credential-based attacks. Recent mitigations strengthen key negotiation, enforce authentication across routing and tunnel layers, and integrate intrusion detection to curb protocol exploits.

Username Enumeration: To mitigate username enumeration in IPsec VPNs using IKEv1 Aggressive Mode, switching to Main Mode—where identity payloads are encrypted—prevents offline guessing [49, 53]. Providers should apply rate limiting, anomaly detection, and strong password policies to block large-scale attempts. Users can reduce exposure by using non-trivial usernames and enabling MFA. Regular firmware and configuration updates are essential to avoid re-introducing known protocol flaws.

Link State Advertisement (LSA) Spoofing: Preventing LSA spoofing in OSPF requires enabling cryptographic authentication (MD5 or SHA-based) to ensure only verified routers exchange LSAs [61]. Administrators should deploy IDS/IPS tools such as Suricata to monitor OSPF traffic, flag abnormal update rates, and drop LSAs from untrusted nodes. Routine key rotation and topology audits further strengthen routing integrity against long-term poisoning attacks.

Adjacent Route Spoofing (Path Spoofing): To counter adjacency spoofing, routers not needing dynamic routing must be set as passive interfaces [61]. Port-check mechanisms should validate legitimate OSPF “hello” packets, while IDS/IPS systems like Suricata can inspect neighbor discovery traffic and block fake adjacency attempts. Periodic review of routing configurations, ACLs, and permissions ensures only authenticated routers form OSPF adjacencies.

Resource Exhaustion via Diffie-Hellman Key Exchanges: Defending against IKE resource-exhaustion attacks requires cookie exchange validation to confirm clients before Diffie-Hellman computation [61]. Implementing rate limits on repeated IKE requests and monitoring UDP 500 traffic with Suricata helps stop

CPU overloads from spoofed handshakes. Multi-threaded IDS setups and regular patching sustain VPN availability and performance under DoS conditions.

Tor: Layered encryption and relay-based routing provide strong anonymity guarantees, yet its architecture remains vulnerable to traffic correlation, malicious relays, and endpoint compromise. Countermeasures proposed in prior work emphasize improving circuit construction policies, relay vetting, and flow obfuscation to resist timing and fingerprinting attacks. Additional focus has been placed on strengthening exit node security and developing adaptive defenses that balance performance with anonymity under active adversarial conditions.

Memory Artifacts in Tor: Tor Browser can leave recoverable volatile and persistent artifacts, including session keys, node descriptors, and visited onion URLs [20, 55]. As of 2021, no complete upstream fix eliminates these traces. Combine engineering controls with user practices: use strict ephemeral key handling, add secure memory scrubbing to the browser runtime, run privacy-focused OSes such as Tails, avoid leaving sessions open, and reboot between sessions to flush RAM. These measures raise the cost for local adversaries but cannot fully prevent memory forensics.

DDoS Attacks: To address volumetric denial-of-service attacks against onion services, Feng and Zhao [21] propose an identity-based signcryption protocol that combines authentication and encryption in a single step. This decentralized trust model replaces reliance on directory authorities with peer-to-peer relay authentication, thereby mitigating Sybil-based flooding and reducing overhead in circuit establishment. The signcryption mechanism ensures that only cryptographically verified relays participate in circuit formation, discouraging malicious node injection and limiting amplification vectors. In addition to reducing handshake latency and authentication bottlenecks, the protocol enforces key freshness, which prevents identity reuse and minimizes exposure to replay-based disruptions.

Web Browsers: Modern web browsers, while increasingly hardened, continue to exhibit exploitable architectural weaknesses. Researchers have proposed a diverse set of mitigation strategies addressing phishing resistance, malware detection, memory management, and browser modularization.

Phishing and Malware: Browser security warnings are a primary defense against phishing and malware [75], but their impact is constrained by user behavior. Akhawe and Felt [38] found that phishing and malware warnings in Chrome and Firefox had low click-through rates (under 25%), whereas over 70% of users ignored SSL certificate alerts. These results expose the limits of passive warning systems and the need for more adaptive, user-aware defenses. To overcome blacklist weaknesses, Virvilis et al. [60] introduced *Secure Proxy*, a network-layer, platform-agnostic filtering system that aggregates antivirus and blacklist data. It enhances protection against malware-hosting sites, unsafe downloads, and insecure mobile browsers without requiring third-party installations.

Mitigation strategies for malware threats fall into three main categories [12, 39]: identifying attack vectors, detecting vulnerable applications, and preventing malicious execution. Wang et al. proposed *HoneyMonkey*, a VM-based system that simulates user browsing to detect drive-by downloads [12]. It filters URLs by

tracking system changes and replays them on vulnerable setups to expose zero-day exploits. Cova et al. developed *JSAND*, a dynamic detection framework combining an instrumented browser with ML classifiers. By analyzing bytecode features such as memory use and redirection depth, JSAND assigns risk scores to web pages, improving detection beyond blacklist-based systems. Additional research employs signature-based detection of stack overflows and component inconsistencies, further enhancing browser-side protection.

Browser Integrity: To address vulnerabilities from residual browser objects and dangling references, Chen et al. [14] advocate implementing engines in memory-safe languages with automatic garbage collection. Although this limits memory corruption, attackers can still block deallocation by maintaining strong references, so memory safety must pair with architectural isolation. A modular, sandboxed browser design confines rendering and scripting compromises, though cross-process mechanisms like COM can weaken isolation if misconfigured [14]. Complementary defenses include systematic model testing of object lifecycles, reference tracking, and navigation behavior [13, 58], which helps detect anomalies and strengthen DOM integrity against untrusted scripts.

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Extensions and Plugins: To counter malicious browser extensions, Toreini et al. [13] proposed *DOMtegrity*, a cryptographic protocol ensuring end-to-end DOM integrity between browser and server. It requires no browser modification or hardware support and uses the WebSocket API for secure two-way communication. Built on the WebExtensions architecture, DOMtegrity isolates extension and page contexts, preventing direct DOM manipulation or message interception. Inline JavaScript instrumentation validates DOM changes, while servers verify integrity reports to block unauthorized modifications or data leaks. Combined with strict API permissioning, this lightweight framework reduces the attack surface created by browser extensibility features.

Browser Design: Grier et al. [40] propose the *OP* browser, which applies OS-style partitioning: a central kernel mediates communication and enforces least-privilege boundaries among rendering, networking, and plugins. Treating each subsystem as untrusted enables fault isolation, forensic tracing, fine-grained access control, and flexible plugin policies, substantially strengthening browser

security. Table 2 summarizes mitigation strategies, mechanisms, and impacts across threat domains.

Table 2: Summary of Mitigation Strategies Across Technologies

Threat Category	Strategy	Mechanism	Measured Impact
Username Enumeration (VPN)	Protocol hardening, rate limiting	Use IKEv1 Main Mode; disable Aggressive Mode; detect anomalies; enforce strong credentials	98% drop in enumeration success (For e.g., [49])
Resource exhaustion	Cookie validation, rate limit	Validate cookies; limit IKE requests; monitor UDP 500 traffic via Suricata	Mitigated DoS, sustained VPN uptime ([61])
LSA Spoofing	Routing auth., IDS/IPS	Enable OSPF MD5/SHA auth.; use Suricata with threshold rules to block spoofed LSAs	Prevented LSA spoofing in VPN tests ([61])
Adjacent Route Spoofing	Passive iface config., port-check	Set unused routers to passive; verify adjacency via port-check; block fake “hello” packets	Stopped route spoofing, stable routing ([61])
Tor Memory Artifacts	Ephemeral key handling, memory wiping	Use Tails OS or live sessions; enforce shutdown; apply secure memory erasure	85–90% fewer recoverable traces (For e.g., [20])
DDoS on Onion Services	Decentralized signcryption-based authentication	Replace directory authority with node-to-node identity verification using signcryption	50–70% reduction in circuit floods (For e.g., [21])
Phishing & Malware (Browser)	Proxy-layer filtering, behavioral detection	Combine blacklists, VirusTotal feeds, and tools like HoneyMonkey, JSAND for dynamic analysis	Up to 92% malware detection; 30% over static methods (For e.g., [12])
Browser Integrity	Safer memory models, modular design	Use garbage-collected languages; sandbox browser subsystems; track object lifecycle	Prevents 60–75% of object leakage/spoofing (For e.g., [14])
Extensions & Plugins	DOM integrity verification	Use DOMtegrity: inline JS + WebSocket to verify DOM state in isolated execution zones	100% detection of malicious DOM changes (For e.g., [13])

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Threat Category	Strategy	Mechanism	Measured Impact
Browser Architecture	Microkernel-style partitioning	Separate core components (render, network, plugins); restrict IPC; enforce info flow control	>85% gain in isolation fidelity (For e.g., [40])

4 Implications

4.1 Cross-Layer Interactions Between Privacy Technologies

Combining privacy tools can boost resilience yet introduce architectural conflicts. VPNs encrypt traffic and mask source IPs at the network layer, while browsers manage identity, sessions, and content at the application layer; this separation can leak information, for example when WebRTC exposes a local IP despite an active VPN [77, 78]. IPsec-based VPNs do not stop application-layer threats such as fingerprinting or malicious scripting [5, 52]. Pairing a hardened browser with a secure tunnel may reduce metadata leakage, but interoperability gaps and misconfigurations often erode privacy guarantees [5, 44, 79, 80]. Building composable privacy stacks requires clear boundaries and coordination across layers: VPNs provide secure transport, Tor anonymizes routing paths, and browsers enforce content integrity [81]. Without alignment, stacking can cause conflicting DNS resolutions, timing correlations, and trust-policy bypasses, allowing fingerprinting to re-identify users despite VPNs and timing to trace Tor traffic [40, 50].

4.2 Security–Usability Trade-offs in Real-World Deployments

Tor’s strict security posture, such as disabling JavaScript by default via NoScript [82], offers strong protection against content-based threats like cross-site scripting and tracking, but at the cost of rendering many websites unusable [40, 43]. While this restriction aligns with Tor’s design goals and threat model, it poses a significant barrier to broader adoption by non-technical users who require access to dynamic content. Mainstream browsers could benefit from adopting more adaptive models that restrict potentially dangerous content only when contextually necessary. For example, selectively isolating third-party scripts or enforcing fine-grained content security policies could offer many of the same benefits as Tor’s default restrictions while preserving a reasonable user experience [44]. However, implementing such models demands coordination among browser vendors, standards bodies, and web developers [50].

4.3 Limitations of VPNs in Application-Layer Threat Models

While VPNs are commonly perceived as comprehensive privacy tools, our findings suggest their protections are far more limited in scope. VPNs primarily

secure transport-level data and obscure source addresses, but they do not shield users from content-based attacks [83–86]. For example, visiting a phishing website or downloading malware remains equally risky with or without a VPN. Because VPNs operate independently of the browser’s content evaluation logic, they cannot verify or sanitize web page contents. Moreover, many VPNs are vulnerable to misconfiguration, IP leaks, or insecure defaults [11, 16, 56]. To improve overall security, VPNs must be used as part of a broader defensive strategy that includes browser hardening, endpoint protection, and user education. VPN clients that integrate with browser APIs to detect anomalies such as WebRTC leaks or DNS prefetching could help bridge the gap between network- and application-layer protections [5, 52]. However, these integrations must be engineered carefully to avoid new attack surfaces or unintended behaviors.

4.4 Designing Context-Aware and Adaptive Privacy Frameworks

The limitations and tensions identified in this review point toward the need for adaptive security frameworks that respond dynamically to risk. Current privacy tools rely heavily on static configurations: Tor disables JavaScript in all contexts, VPNs encrypt all traffic indiscriminately, and browsers enforce permissions uniformly across sites. While effective in constrained threat environments, such rigid configurations often fail to balance performance, usability, and privacy in more nuanced or evolving contexts. Future systems should explore the integration of context-aware models that adjust protections based on environmental cues, user behavior, or site reputation. For example, a browser could selectively isolate or disable untrusted content when visiting unknown domains, while relaxing restrictions for known-safe sites. Similarly, VPNs could offer risk-based routing modes that adapt tunnel parameters based on detected threats or data sensitivity. These adaptive mechanisms must remain interpretable, auditable, and under user control to avoid compromising trust or privacy [5, 44, 87].

5 Future Work and Limitations

We focused our review on peer-reviewed academic sources, which may not reflect the full scope of emerging vulnerabilities disclosed in industry or grey literature. In future work, we plan to expand our corpus to capture overlooked threat vectors and evolving adversarial techniques. We also plan to conduct empirical user studies to examine how individuals adopt and apply proposed defenses in practice. To complement our synthesis, we will perform system-level experiments to evaluate the effectiveness, deployability, and resilience of selected mitigation strategies under realistic operating conditions.

6 Conclusion

As users increasingly adopt privacy-enhancing technologies to defend against online threats, it becomes essential to understand the interaction boundaries, security limitations, and failure modes of systems like VPNs, and the Tor network. In

this study, we conducted a systematic review of prior research addressing vulnerabilities and mitigation strategies across these platforms. Our findings show that no single tool offers complete protection in isolation. Web browsers remain exposed to client-side threats such as JavaScript-based fingerprinting, DOM-based data leaks, and malicious extensions, even when paired with VPNs. While VPNs effectively secure transport-layer metadata, they fail to prevent application-layer attacks or endpoint compromise. Tor enhances anonymity through onion routing and restrictive browser configurations, but often does so at the cost of usability and compatibility with dynamic web content. We identified several promising defense mechanisms, such as DOMtegrity, JSAND, and identity-based signcryption that reduce attack surfaces while preserving performance. However, these approaches face barriers to adoption due to integration complexity and lack of mainstream support. Our review also revealed gaps in the literature, including underexamined memory artifacts in Tor and username enumeration vulnerabilities in VPN authentication flows. These findings highlight the need for cross-layer, context-aware security architectures that coordinate transport encryption, content isolation, and adaptive risk response.

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